Tips for Preparing Methods, Statistics, and Results Sections

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- Statistical Editor, *Headache*

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- StatReviewer Software
  - Co-founder

Outline

- General tips for submission to *Anesthesiology*
- Preparing a methods section
- Preparing a statistical analysis section
- Preparing a results section

General Tips for Submission to *Anesthesiology*

- Read the instructions for authors
  - https://pubs.asahq.org/anesthesiology/pages/instructions-for-authors
- IMRAD Format
  - Structured abstract
  - Statistical analysis section

Anesthesiology reviewer ratings

Clinical Impact (1 = None, 3 = Average, 5 = Immense)
Definitive (1 = Not at all, 3 = Average, 5 = Completely) ✓
Interesting to the Specialty (1 = Not at all, 3 = Average, 5 = Completely)
Novelty (1 = Not at all, 3 = Average, 5 = Completely)
Scientific Impact (1 = None, 3 = Average, 5 = Immense) ✓

Strategy for Methods Preparation

- Attend to ethical issues
  - Ethical board approval
  - Consent/Assent
- Avoid ‘Salami’ publishing
- Be careful about duplicate publication
- Name the study design
- Use checklist to report the methods

Carefully attend to ethical issues

- Ethical violations arise for a variety of reasons:
  - Commercial gain
  - Academic advancement
  - Intentional disregard of ethical principles
  - Ignorance of ethical standards
- Implications of ethical misconduct:
  - Serious repercussions/career implications for authors
  - Diminish validity and value of scientific literature
  - Undermine public trust
- Clearly report the ethical considerations underlying the research
  - Ethical board approval
  - Written informed consent

Avoid redundant or 'Salami' publishing

- Redundant publishing: Multiple papers with same design, methods, data, outcomes, & conclusions

- Publishing findings reported at meetings or on trial registrations sites not deemed 'duplicate'

- Direct translation from another language *may or may not* be unethical

Be careful about duplicate publication

- Plagiarism well recognized as unethical
- Self- or auto-plagiarism less familiar concept – substantial overlap of content between 2 or more papers by the same author
- To an extent, repetition is legitimate and unavoidable (e.g., methods section)
- Problematic when overdone – common examples:
  - Duplication of findings within results section
  - Re-use of table or figure without proper citation (copyright infringement)

Name the Design

- Is it clear what type of study was conducted?
  - Title
  - Abstract
  - Methods
- Can the stated aims/hypotheses be addressed given the design?
- What are the inherent limitations in the design?
- What type of analyses will be needed given the design?

Name the Design

- All empirical research can be placed into one study design type
  - Clinical Trial
  - Observational (Cohort, Case-Control, Longitudinal)
  - Experimental Study (Animal)
  - Systematic Review/Meta-analysis
  - Diagnostic Accuracy Study

- Different systems for categorizing research.

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Library for health research reporting

The Library contains a comprehensive searchable database of reporting guidelines and also links to other resources relevant to research reporting.

- Search for reporting guidelines
- Not sure which reporting guideline to use?
- Reporting guidelines under development
- Visit the library for more resources

Reporting guidelines for main study types

Randomised trials	CONSORT	Extensions
Observational studies	STROBE	Extensions
Systematic reviews	PRISMA	Extensions
Study protocols	SPIRIT	PRISMA-P
Diagnostic/prognostic studies	STARD	TRIPOD
Case reports	CARE	Extensions
Clinical practice guidelines	AGREE	RIGHT
Qualitative research	SRQR	COREQ
Animal pre-clinical studies	ARRIVE	
Quality improvement studies	SQUIRE	
Economic evaluations	CHEERS	

See all 405 reporting guidelines

Strategy for Statistical Analysis Preparation

- Sufficient details
- Avoid THE stats section
- Attend to level of measurement
- Statistical power considerations

Statistical Analyses Details

- Provide enough detail to allow replication if allowed access to your data:
  - Statistical Software
  - Criterion for statistical significance ($p < 0.05$)
  - Hypothesis testing (two-tailed)
  - Assumptions underlying tests (normality, etc.)
  - Actual applied tests, not just a list
  - Reporting of data (mean, median, etc.)
  - Correction for multiplicity

Avoid *THE* stats section

Statistical analysis

Categorical variables are reported as count (%) and continuous variables as mean (SD) or median (IQR). The presence of normal distribution was verified by Kolmogorov-Smirnoff test. We used the *t* test to assess differences between parametric continuous variables, the Mann-Whitney *U* test for non-parametric variables, the χ^2 test for categorical variables, and the Fisher exact test for 2×2 tables. No correction for multiple testing was done. A two-sided $p<0{\cdot}05$ was considered statistically significant. All analyses were done with SPSS version 24.0.

https://twitter.com/MaartenvSmeden/status/1443898256187088896

Type of Measurements

- How are the ideas quantified?
  - Have these measurements been validated previously?

- Do the measurements correspond to the ideas being studied (i.e., validity)?

- Would these measurements be expected to be reproducible (i.e., reliable)?

Level of Measurement

+ Information -

<u>Nominal</u>

Binary: dead/alive, yes/no

Categorical: ethnicity, medical diagnosis

<u>Ordinal</u>: ASA, morbidity scale, attitudes, rating scales

<u>Interval</u>: IQ, Pain ratings

<u>Ratio</u>: Age, walking speed, weight

Failure to conduct and/or report a statistical power analysis

- Power analyses are very important for interpreting null findings (positive findings, too)

- Without power, cannot distinguish between the lack of a ‘true’ effect versus the lack of power to detect one
  - Clinically meaningful difference specified

- Post hoc power analyses are misleading

Strategy for Results Preparation

- Missing data
- Results reporting
- Interpretation/ Spin

Missing data (or lost data)

- Missing data pose a threat to the interpretation of all studies

- Data not collected for all individuals
    - Only those not in acute distress
    - Only those with a certain hospital length of stay
    - Only those animals that survived experimentation

  – Biomarker assay not available for all levels
    - Minimal detectable threshold

Missing data in submitted manuscripts

What to do about missing data?

- Must know that data are "missing"
  - Reporting of protocol is crucial
    - What patient groups are not being included?
- Characterization of missing data
  - For whom, when, and theoretical effects
  - Standard missing value analysis
- Then, estimation of bias is possible
  - Multiple imputation, clustering methods, etc.
  - Secondary to experimental design (the best way to handle missing data is to not have it)

Reporting of Results

- Can you identify the size of the effects?
  – Are more than just p-values reported?
  – Descriptive statistics!

- What is the uncertainty around the findings?
  – Not just a dichotomous finding (statistically significant vs. not)
  – Variability estimates (SD, IQR[25^{th}, 75^{th}])
  – Confidence intervals!

OR 3.5 (95%CI: 3.0 to 4.0)

Common Error: Worshipping the p-value

- Solitary p-value
  - "The control group was inferior to the treatment group on mortality ($p < 0.05$), length of stay ($p < 0.05$), and adverse events ($p < 0.05$)"
- Contrasting p-values
  - "The treatment group was significantly different than baseline ($p < 0.05$), but the control group was not ($p > 0.05$)"
- Failed Superiority ≠ Equivalence
  - "The groups were similar on the primary outcome, $p = 0.745$"

https://www.elsevier.com/__data/promis_misc/ejvesstat.pdf

Interpretation

- Does the interpretation reflect the actual data?
  - Primary outcomes treated as primary?
  - Focus on significant findings while ignoring non-significant ones

- Can the study design support the interpretations?
  - Convenience sampling cannot yield population prevalence estimates
  - Observational studies cannot be interpreted as randomized designs
  - Beware the pre-test post-test design!

Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

Boutron I, Dutton S, Ravaud P, Altman D. (May, 2010)

- Spin[†]: specific reporting that could distort the interpretation of results and mislead readers.
  - From ignorance of scientific (statistical) issue
  - Unconscious bias
  - Willful intent to deceive

†Fletcher RH, Black B. "Spin" in scientific writing: scientific mischief and legal jeopardy. *Med Law. 2007;* 26(3):511-525.

Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

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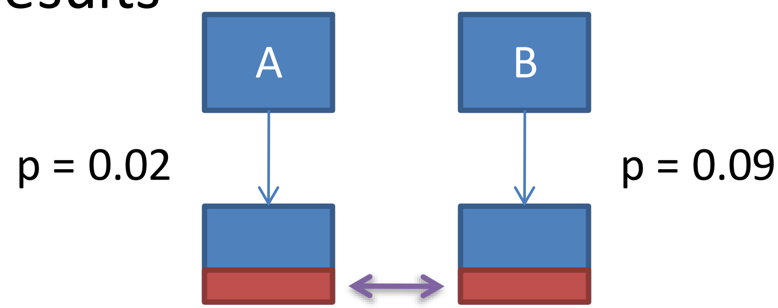
Strategies of Spin

- A focus on statistically significant results
  - within-group comparison
  - secondary outcomes
  - subgroup analyses
  - modified population of analyses



- Interpreting statistically nonsignificant results for the primary outcomes as showing treatment equivalence or comparable effectiveness

- Claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results.

Final Take Home Messages

- Identify the study design, state it, and focus on the reporting required to understand the important methods related to it

- Provide enough details to allow replication of the statistical analyses

- Report the findings with a wealth of descriptive statistics

- Report point estimates of effect (e.g., group differences) and 95%CI of those effects.

Thank You!

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